

SAO LUIS, Brazil

WMO: 822810

Lat: 2.585S

Lon: 44.234W

Elev: 54

StdP: 100.67

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	22.8	23.1	17.8	12.8	31.2	18.5	13.4	30.6	8.3	28.9	7.5	29.4	2.1	60	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	6.4	33.3	25.5	33.0	25.4	32.2	25.0	27.6	30.4	27.2	30.2	26.8	29.9	6.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.8	29.3	26.2	21.8	28.5	26.1	21.6	28.4	88.4	30.3	86.3	30.4	84.4	30.5	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.5	7.9	DB	21.2	34.9	2.3	1.4	19.5	35.9	18.2	36.7	17.0	37.5	15.3	38.5	(4)
(5)			WB	19.9	28.1	1.9	1.7	18.6	29.3	17.5	30.3	16.4	31.3	15.1	32.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	27.9	27.5	27.4	27.4	27.8	27.9	27.9	28.4	28.7	29.0	29.1	28.9
	DBStd	1.15	1.24	1.32	1.18	1.07	1.00	0.82	0.85	0.74	0.70	0.64	0.69	1.08
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6628	556	490	538	521	552	537	554	569	562	589	573	586
	CDD18.3	3587	297	257	280	271	294	287	296	311	312	331	323	328
	CDH23.3	36479	3035	2451	2577	2419	2787	2798	2869	3178	3338	3663	3654	3710
	CDH26.7	12088	968	718	697	648	847	896	923	1086	1183	1344	1378	1401
Wind	WSAvg	3.9	4.0	3.4	3.0	2.5	2.6	2.8	3.2	4.2	5.1	5.5	5.5	5.1
Precipitation	PrecAvg	1787	187	256	326	357	267	147	105	30	8	5	10	61
	PrecMax	2678	450	482	577	602	539	262	244	82	35	53	40	264
	PrecMin	995	18	95	123	156	78	23	26	8	1	0	1	21
	PrecStd	435	125	120	109	136	113	61	50	20	7	10	9	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	33.0	32.3	32.4	33.0	32.9	33.1	33.2	34.0	34.0	34.1	34.1
		MCWB	25.8	25.8	26.0	26.0	26.3	25.6	25.3	25.0	25.7	25.6	25.9	26.4
	2%	DB	32.2	32.0	31.5	31.8	32.0	32.1	32.1	32.6	32.9	33.0	33.0	33.0
		MCWB	25.2	25.4	25.9	25.9	25.6	25.1	24.9	24.9	24.9	25.0	25.4	25.6
	5%	DB	31.4	31.1	30.9	30.9	31.2	31.2	31.2	31.9	32.1	32.1	32.2	32.2
		MCWB	25.0	25.4	25.7	25.9	25.7	25.1	24.9	24.8	24.6	24.7	24.8	24.9
	10%	DB	30.8	30.2	30.0	30.0	30.7	30.8	30.8	31.1	31.4	31.8	31.9	31.8
		MCWB	24.9	25.3	25.5	25.7	25.6	25.1	24.8	24.8	24.5	24.5	24.7	24.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.8	27.7	28.0	28.1	27.6	27.5	27.2	26.7	26.9	27.1	27.5
		MCDB	30.2	30.2	29.9	30.3	30.7	30.5	30.3	30.7	31.1	31.1	31.5	31.3
	2%	WB	26.9	27.1	26.9	27.1	27.2	26.8	26.8	26.6	26.3	26.3	26.6	26.8
		MCDB	29.7	29.6	29.7	29.8	30.2	30.0	29.8	30.2	30.5	30.5	30.7	30.1
	5%	WB	26.4	26.5	26.3	26.4	26.6	26.2	26.2	26.1	25.9	26.0	26.2	26.5
		MCDB	29.3	29.0	29.1	29.1	29.5	29.3	29.3	29.8	30.1	30.1	30.1	29.9
	10%	WB	26.0	26.1	26.0	26.0	26.1	25.9	25.8	25.7	25.5	25.6	25.9	26.1
		MCDB	29.0	28.6	28.7	28.7	28.9	28.9	28.9	29.2	29.4	29.5	29.6	29.7
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.0	6.0	6.0	6.4	6.7	6.7	6.9	6.7	6.6	6.4	6.2
		MCDBR	6.6	6.7	6.7	6.9	6.9	7.1	7.2	7.4	7.2	6.8	6.7	6.6
		MCWBR	2.2	2.4	2.7	2.8	2.6	2.5	2.5	2.4	2.1	2.0	2.0	1.9
	5% WB	MCDBR	6.2	6.1	6.2	6.3	6.6	6.9	6.9	7.2	7.1	6.8	6.8	6.5
		MCWBR	2.7	2.8	2.8	2.9	2.9	2.8	2.7	2.6	2.4	2.3	2.3	2.3
Clear-Sky Solar Irradiance	taub	0.427	0.417	0.417	0.413	0.414	0.403	0.411	0.446	0.510	0.496	0.489	0.466	
	taud	2.448	2.491	2.498	2.528	2.505	2.533	2.493	2.363	2.134	2.186	2.212	2.303	
	Ebn at Noon	908	918	908	889	863	860	858	848	815	837	844	866	
	Edh at Noon	119	115	113	106	105	100	105	124	161	154	150	136	
All-Sky Solar Radiation	RadAvg	4.95	4.77	4.69	4.63	4.82	5.09	5.29	5.84	6.05	5.89	5.68	5.33	
	RadStd	0.34	0.42	0.32	0.28	0.34	0.20	0.18	0.12	0.18	0.17	0.18	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-1.84	-0.50	-0.66	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page